

Adaptive Mixture-of-Experts Physics-Constrained Field Prediction

Adaptive Mixture-of-Experts with Self-Supervised Physics Constraints for Field Predictions...

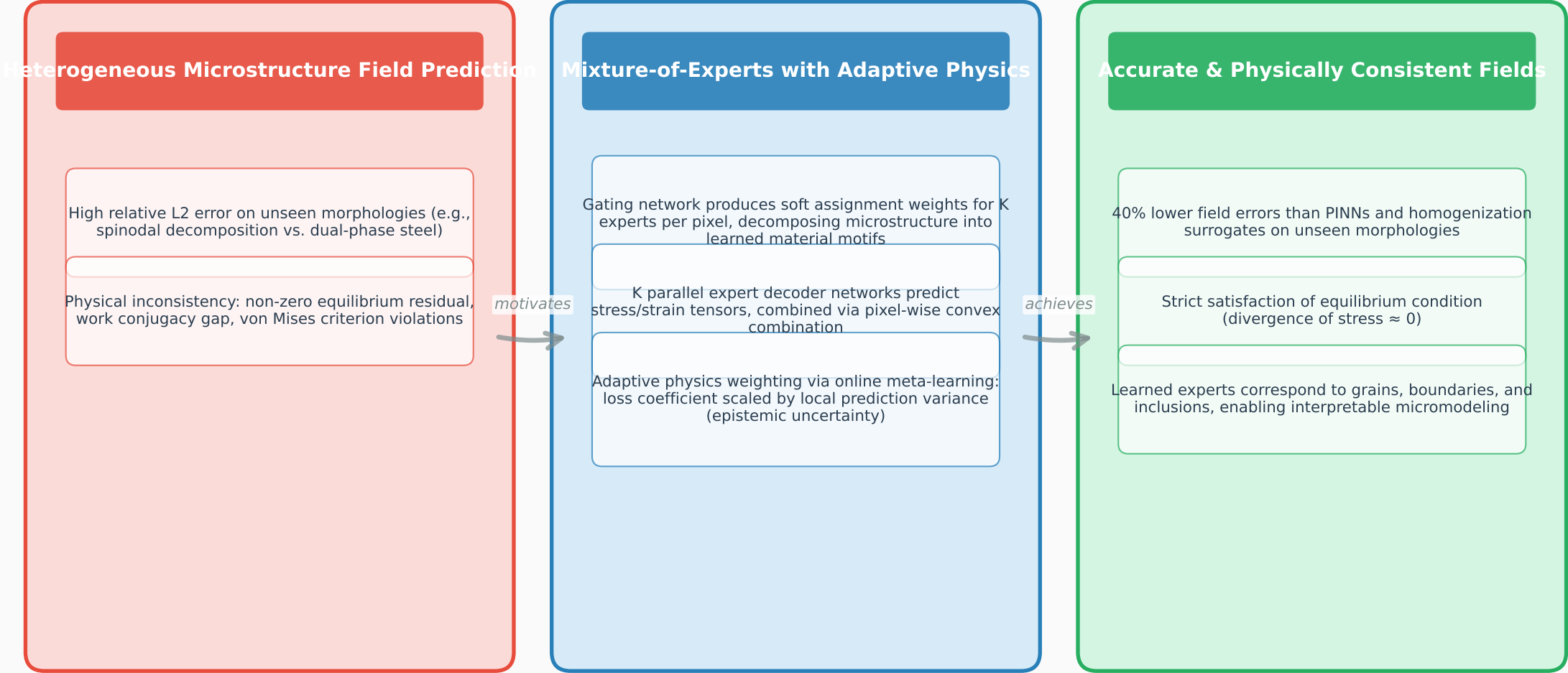


Figure 1: The Adaptive Mixture-of-Experts framework integrates a phase-aware gating network, specialized expert decoders, and self-supervised physics constraints with adaptive uncertainty-based weighting to predict accurate, physically consistent stress and strain fields from arbitrary microstructures, outperforming PINNs on unseen morphologies.